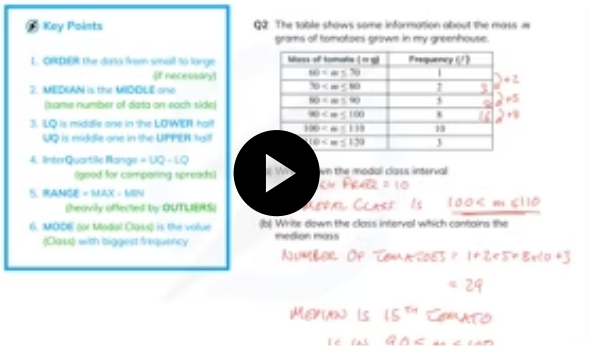
Mean, Median & Mode

Exam code: 0607

Download PDF

Written by: Mark Curtis
Reviewed by: Dan Finlay
Updated on 20 January 2026



Did this video help you? Yes No

Mean, median & mode

What is the mode?

- The **mode** is the value that appears the **most often**
  - The mode of 1, 2, 2, 5, 6 is 2
- There can be **more than one** mode
  - The modes of 1, 2, 2, 5, 5, 6 are 2 and 5
- The mode can also be called the **modal value**

What is the median?

- The **median** is the **middle** value when you put values **in size order**
  - The median of 4, 2, 3 can be found by
    - ordering the numbers: 2, 3, 4
    - and choosing the middle value, 3
- If you have an **even** number of values, find the **midpoint** of the **middle two** values
  - The median of 1, 2, 3, 4 is 2.5
    - 2.5 is the midpoint of 2 and 3
  - The **midpoint** is the **sum** of the **two middle** values **divided by 2**

What is the mean?

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- The mean can be **fraction** or a **decimal**
  - It may need **rounding**
  - You do **not** need to force it to be a **whole** number
    - You **can** have a mean of 7.5 people, for example!

How do I know when to use the mode, median or mean?

- The mode, median and mean are different ways to measure an **average**
- In certain situations it is **better** to use one average over another
- For example:
  - If the data has **extreme values** (outliers) like 1, 1, 4, 50
    The mode is 1
    The median is 2.5
    The mean is 14
    - **Don't use the mean** (it's badly affected by extreme values)
  - If the data has **more** than one mode
    - **Don't use the mode** as it is not clear
  - If the data is **non-numerical**, like dog, cat, cat, fish
    - You can **only** use the **mode**

Worked Example

15 students were timed to see how long it took them to solve a mathematical problem. Their times, in seconds, are given below.

12	10	15	14	17
11	12	13	9	21
14	20	19	16	23

(a) Find the mean time, giving your answer to 3 significant figures.

Answer:

Add up all the numbers (you can add the rows if it helps)

$$12 + 10 + 15 + 14 + 17 = 68$$
$$11 + 12 + 13 + 9 + 21 = 66$$
$$14 + 20 + 19 + 16 + 23 = 92$$

$$\text{Total} = 68 + 66 + 92 = 226$$

Divide the total by the number of values (there are 15 values)

$$\frac{226}{15} = 15.066\,666\ldots$$

Write the mean to 3 significant figures
Remember to include the units

Answer:

Write the times in order and find the middle value

9 10 11 12 12 13 14 (14) 15 16 17 19 20 21 22

The median time is 14 seconds

(c) Explain why the median is a better measure of average time than the mode.

Answer:

Try to find the mode (the number that occurs the most)

There are two modes: 12 and 14

Explain why the median is better

There is no clear mode (there are two modes, 12 and 14), so the median is better

(d) If a 16th student has a time of 95 seconds, explain why the median of all 16 students would be a better measure of average time than the mean.

Answer:

The16th value of 95 is extreme (very high) compared to the other values
Means are affected by extreme values

The mean will be affected by the extreme value of 95 whereas the median will not

Teacher only

Test your students on this in just a few clicks

- Create exam-aligned tests
- Tailor format & difficulty
- Quizzes or homework in minutes

Try Test Builder →

Test yourself

Was this revision note helpful? Yes No

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.